

Postpartum Medicaid Extensions, Outpatient LARC Utilization, and the Medicaid Unwinding: Multi-Channel Evidence from Pharmacy and Claims Data

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Word Count: ~10,000 (excluding tables, figures, and references)

Keywords: Medicaid, postpartum coverage extension, LARC, contraceptive access, difference-in-differences, Medicaid unwinding

Funding: None.

Conflicts of Interest: The authors report no conflicts of interest.

Abstract

Background. Section 9812 of the American Rescue Plan Act of 2021 gave states the option to extend Medicaid postpartum coverage from 60 days to 12 months. By late 2024, 46 states and the District of Columbia had adopted this extension, yet nationally representative causal evidence on outpatient contraceptive utilization remains limited, and the concurrent Medicaid unwinding complicates policy evaluation during the same period.

Methods. We constructed a balanced state-quarter panel of 51 jurisdictions over 36 quarters ($N = 1,836$) combining Medicaid State Drug Utilization Data (SDUD), which capture pharmacy-dispensed prescriptions, with DOGE T-MSIS outpatient claims, which capture HCPCS- and CPT-billed LARC procedures. We merged both sources with CMS Medicaid Budget and Expenditure System enrollment data. Exploiting staggered state adoption of the 12-month extension, we estimated two-way fixed effects (TWFE), Gardner (2022) two-stage difference-in-differences (did2s), and local projections difference-in-differences (LPDID) models across pharmacy, outpatient procedure, and combined channels. We then used extended drug-category analyses to diagnose the unwinding confound and re-estimated triple-difference (DDD) specifications on transformed relative scales using actual negative control drugs (statins, BPH medications, gout medications, and Alzheimer’s drugs).

Results. Standard estimates showed no detectable outpatient LARC effect in any channel: the combined estimate was -0.055 per 1,000 enrollees ($p = 0.604$), the pharmacy-only estimate was -0.010 ($p = 0.915$), and the outpatient-procedure estimate was -0.119 ($p = 0.140$). Negative control drugs with no postpartum clinical connection nonetheless showed negative TWFE coefficients, indicating that the 2023-2024 Medicaid unwinding acts as a broad confound. After re-estimating the LARC-specific DDD on a relative scale using actual negative controls, the primary pooled estimate was modestly positive but imprecise (TWFE log1p DDD = +0.094, $p = 0.095$; did2s = +0.109, $p = 0.078$). Alternative normalizations attenuated the estimate further. Power analysis confirmed that

the raw TWFE design could not detect effects of the magnitude found by Rodriguez et al. (2024) after denominator dilution (MDE = 0.255 vs. benchmark = 0.185 per 1,000 total enrollees).

Conclusions. The strongest evidence in this paper is not a definitive positive first stage, but a disciplined combination of findings: outpatient LARC estimates are null across pharmacy and claims channels, the Medicaid unwinding materially distorts raw post-2022 comparisons, and scale-adjusted DDD specifications provide only suggestive evidence of a positive relative LARC response. More definitive evidence on total postpartum contraceptive uptake likely requires inpatient IPP-LARC claims and postpartum-specific denominators.

1. Introduction

1.1 The Postpartum Coverage Crisis

Maternal mortality in the United States is among the highest in the developed world, and unlike most peer nations, the rate has been rising. Between 2018 and 2021, the maternal mortality rate increased from 17.4 to 32.9 deaths per 100,000 live births, with stark racial disparities: non-Hispanic Black women experienced mortality rates approximately 2.6 times those of non-Hispanic White women (NCHS, 2023). One-third of pregnancy-related deaths occur between one week and one year postpartum, yet federal law historically required states to provide pregnancy-related Medicaid coverage through only 60 days after delivery (MACPAC, 2023). This temporal mismatch – between the period of greatest medical vulnerability and the period of guaranteed coverage – has been the subject of sustained policy attention.

Medicaid is the single largest payer for maternity care in the United States, financing approximately 42% of all births (Martin et al., 2023). For low-income women, Medicaid often provides the only pathway to comprehensive prenatal, delivery, and postpartum services, including contraceptive care. The program’s pregnancy-related eligibility pathway typically extends coverage to women with incomes up to 138-200% of the federal poverty level, but this eligibility has historically terminated 60 days after delivery, creating the “coverage cliff.”

The consequences of this coverage cliff are well documented. Daw, Hatfield, Swartz, and Sommers (2017) provided the first nationally representative evidence of postpartum coverage disruption, showing that 55% of women with Medicaid or CHIP coverage at delivery experienced a coverage gap within six months of childbirth. This churn is not merely an administrative inconvenience; it disrupts continuity of care during a period of elevated risks for postpartum depression, cardiovascular complications, substance use relapse, and unintended pregnancy (Gordon et al., 2020). Gordon, Sommers, Wilson, and Trivedi (2020) used a difference-in-differences design comparing Colorado to Utah and found that ACA Medicaid expansion yielded approximately 0.9 additional months of postpartum coverage and 0.52 additional outpatient visits, with larger gains among women with serious delivery complications. Their findings established that broader Medicaid eligibility could mitigate the coverage cliff, though the gains were modest and incomplete.

1.2 Short Interpregnancy Intervals and Contraceptive Access

Short interpregnancy intervals (IPI) – defined as less than 18 months between delivery and subsequent conception – are associated with significantly increased risks of preterm birth, low birth weight, and small-for-gestational-age births. The meta-analysis by Conde-Agudelo, Rosas-Bermudez, and Kafury-Goeta (2006) analyzed 67 studies and confirmed these associations.

ACOG’s 2019 Obstetric Care Consensus endorsed intervals of at least 18 months and recommended that clinicians address contraception beginning in the postpartum period (ACOG, 2019).

Racial disparities in short IPI are pronounced: intervals of less than 6 months are most common among non-Hispanic Black mothers (7.1%), compared to Hispanic (5.0%) and non-Hispanic White (4.1%) mothers (NCHS, 2016). These disparities are partly driven by differential access to postpartum contraception. Thiel de Bocanegra, Braughton, Bradsberry, and colleagues (2017) examined 199,860 California Medicaid births and found that Black women were less likely to receive postpartum care, any contraception, and highly effective contraception compared to White women. The coverage cliff therefore disproportionately affects the populations that already face the greatest barriers to adequate birth spacing.

Long-acting reversible contraceptives (LARC), including intrauterine devices (IUDs) and subdermal implants, are among the most effective methods for preventing unintended pregnancy and extending interpregnancy intervals. LARC methods have failure rates below 1% in typical use, compared with approximately 7% for oral contraceptives and 13% for male condoms (ACOG, 2019). Liberty, Yee, Darney, Lopez-Defede, and Rodriguez (2020) used linked South Carolina Medicaid claims and birth certificate data on 187,438 births and found that receipt of immediate postpartum LARC was associated with decreased odds of a subsequent short IPI (aOR 0.62, 95% CI 0.44-0.89), providing individual-level evidence of the LARC-IPI pathway.

1.3 The ARPA Section 9812 Option

Section 9812 of the American Rescue Plan Act (ARPA) of 2021 created a state plan amendment (SPA) option allowing states to extend Medicaid postpartum coverage from 60 days to 12 months, effective April 1, 2022. CMS released implementation guidance in December 2021 (SHO #21-007), and the Consolidated Appropriations Act of 2023 made this option permanent. By late 2024, 46 states and the District of Columbia had adopted the extension, with Illinois implementing as early as July 2021 via an 1115 waiver and subsequent adoption concentrated in 2022-2023 (KFF, 2024). MACPAC recommended that Congress mandate the extension for all states with 100% federal matching.

This staggered adoption across states and time creates a natural experiment suitable for difference-in-differences identification. The variation in adoption timing reflects a combination of political will, administrative capacity, and pre-existing Medicaid program structure rather than differential trends in contraceptive utilization. However, the adoption window coincides with two major policy shocks – the end of the COVID-19 continuous enrollment provision and the *Dobbs v. Jackson Women’s Health Organization* decision – both of which complicate causal inference.

1.4 The Research Gap

Despite rapid policy uptake and its potential implications for maternal and infant health, no study has estimated the causal effect of 12-month postpartum Medicaid coverage extensions on pharmacy-dispensed medication utilization using nationally representative data. Existing evidence addresses related but distinct policies. Steenland, Pace, Sinaiko, and Cohen (2019, 2021) evaluated separate Medicaid reimbursement for immediate postpartum LARC (IPP-LARC) in South Carolina and found increased uptake and decreased short-interval births among adolescents. Rodriguez, Meath, Watson, and colleagues (2024) published the largest multi-state IPP-LARC study in *JAMA Health Forum*, using T-MSIS data on 1,378,885 deliveries across 15 states and finding a 0.74 percentage-point increase in IPP-LARC use. Eliason, Spishak-Thomas, and Steenland (2022) found a 7.0

percentage-point increase in postpartum LARC use from ACA Medicaid expansion. Swartz et al. (2025) used individual-level North Carolina Medicaid data to show increased postpartum contraceptive utilization after the extension, but this single-state pre-post comparison cannot isolate the causal effect.

The 12-month postpartum extension is a different policy lever from any of these: it extends the *duration* of eligibility rather than reforming reimbursement rules (as with IPP-LARC policies) or expanding *initial* eligibility (as with ACA expansion). Several null findings temper expectations. Pace, Steenland, Engel, and Sinaiko (2021) found no significant per-capita LARC effect from Medicaid expansion despite absolute increases in prescriptions. Law et al. (2016) found no immediate LARC uptake increase after the ACA contraceptive mandate. These findings suggest that coverage alone may be insufficient when supply-side barriers – including provider training, device stocking, and scheduling logistics – persist.

1.5 Contributions

This paper makes four contributions to the literature on Medicaid coverage policy and postpartum contraceptive access.

First, we develop a multi-data-source framework combining Medicaid State Drug Utilization Data (SDUD) and DOGE T-MSIS outpatient claims to assess postpartum medication utilization across pharmacy and outpatient procedure channels. The claims upgrade materially improves measurement of outpatient LARC procedures and yields a cleaner test of whether the raw SDUD null is merely a missing-channel artifact. To our knowledge, this is the first study to combine these data sources for a contraceptive policy evaluation, and the multi-channel framework is portable to other Medicaid drug utilization studies.

Second, we identify the 2023-2024 Medicaid unwinding as a critical confound in contemporaneous Medicaid policy evaluations. Negative control drugs with no postpartum clinical relevance show negative TWFE coefficients, indicating that raw post-2022 contrasts are contaminated by a broad enrollment shock. This finding has implications well beyond the current study: any difference-in-differences evaluation of Medicaid policies adopted during 2022-2024 must contend with the unwinding as a simultaneous shock.

Third, once the LARC-specific triple-difference (DDD) is re-estimated on a defensible relative scale using the paper’s actual negative controls, the positive signal becomes suggestive rather than definitive (TWFE log1p DDD: +0.094, $p = 0.095$; did2s: +0.109, $p = 0.078$), and alternative normalizations attenuate further. The strongest causal claim is therefore not that the extension clearly increases outpatient LARC utilization, but that raw nulls are partly confounded while the remaining positive evidence is modest and sensitive to specification.

Fourth, formal power analysis confirms that the raw TWFE design cannot detect effects of the magnitude found by Rodriguez et al. (2024) after denominator dilution: the minimum detectable effect for the pharmacy channel is 0.255 per 1,000 total enrollees (26% of baseline), which exceeds the Rodriguez benchmark of 0.185. Together, these results imply that stronger first-stage evidence likely requires inpatient IPP-LARC claims or postpartum-specific denominators.

Our findings carry immediate implications for researchers and policymakers. The identification of the unwinding as a pervasive confound – evidenced by significant negative coefficients for negative control drugs with no postpartum relevance – provides a cautionary finding for all contemporaneous Medicaid policy evaluations and a methodological template for diagnosing common shocks during

the redetermination period. The re-estimated LARC DDDs are directionally positive but modest and imprecise enough that they should be treated as supportive relative evidence rather than proof of a large first stage. Back-of-envelope denominator adjustment illustrates the scale of attenuation: postpartum-eligible women represent approximately 1.9% of total Medicaid enrollment, yielding a scaling factor of approximately 57x. The broader reduced-form analysis of interpregnancy intervals still awaits restricted NVSS natality data.

The remainder of this chapter proceeds as follows. Section 2 provides institutional background and reviews the relevant literature. Section 3 describes the data sources and sample construction. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 discusses findings, and Section 7 concludes.

2. Background and Literature Review

2.1 The Postpartum Coverage Cliff and Its Consequences

The foundational policy problem motivating this study is the historically limited duration of pregnancy-related Medicaid coverage. Under federal law prior to ARPA, states were required to provide Medicaid coverage for pregnant women during pregnancy and through 60 days postpartum, after which coverage terminated unless the woman qualified through another eligibility pathway (e.g., adult expansion, disability). The 60-day cutoff was established by the Omnibus Budget Reconciliation Act of 1986 and had not been updated for over three decades despite substantial changes in the understanding of postpartum health needs.

The clinical rationale for longer postpartum coverage is compelling. ACOG has emphasized that the postpartum period is increasingly understood as extending to 12 months, with ongoing needs for chronic disease management, mental health treatment, substance use disorder care, and contraceptive services (ACOG, 2018). The CDC’s Pregnancy Mortality Surveillance System documented that approximately one-third of pregnancy-related deaths occur between 43 days and one year postpartum, falling entirely outside the guaranteed coverage window under the 60-day rule.

Daw et al. (2017) provided the first nationally representative quantification of the coverage cliff’s consequences. Using longitudinal MEPS data, they showed that coverage churn around pregnancy was pervasive: 55% of women with public insurance at delivery experienced a gap within six months. The coverage disruption was not randomly distributed; it disproportionately affected women of color, women with lower educational attainment, and women in states that had not expanded Medicaid under the ACA. This differential exposure to coverage loss has important implications for health equity, as the populations most likely to lose coverage are also those with the highest rates of adverse postpartum outcomes. Johnston, McMorrow, Thomas, and Kenney (2020) similarly documented that Medicaid expansion reduced uninsurance among new mothers in poverty, but significant coverage gaps persisted.

The COVID-19 pandemic created a natural experiment that powerfully demonstrated the magnitude of coverage losses under the 60-day rule. The Families First Coronavirus Response Act (FFCRA) continuous enrollment provision increased the rate of continuous 12-month postpartum Medicaid enrollment from 59.3% to 90.7%, corresponding to approximately 430,000 fewer people losing postpartum coverage (Gordon et al., 2024). This finding revealed that roughly 40% of Medicaid-covered births previously resulted in coverage loss before 12 months postpartum – a staggering rate of disruption that the 12-month extension was designed to address.

2.2 Section 9812 and State Adoption

Section 9812 represented the most significant legislative change to postpartum Medicaid coverage since the original 60-day mandate. CMS implementation guidance (SHO #21-007) specified that the extension applied to all pregnancy-related eligibility categories and covered the full Medicaid benefit package (not just pregnancy-related services). State adoption was remarkably rapid by the standards of optional Medicaid provisions: the modal adoption quarters were 2022 Q3 and 2022 Q4, each with 9 states, and by late 2024 only five jurisdictions (Arkansas, Florida, New Jersey, Virginia, and Wisconsin) had not yet implemented (KFF, 2024). Some states, most notably Illinois, implemented earlier through Section 1115 waivers. The compressed adoption window is important for identification: it provides substantial cross-state variation in treatment timing but limits the power of heterogeneous treatment effect analyses by adoption cohort. ASPE issue briefs estimated that hundreds of thousands of additional person-months of coverage are generated annually by the extension (Gordon et al., 2023).

The mechanism through which the extension could affect contraceptive utilization operates through the demand side: by extending the period of guaranteed insurance coverage, the extension reduces the effective price of contraceptive services for postpartum women who would otherwise lose coverage at 60 days. For LARC specifically, the extension may matter most for women who do not receive LARC during the delivery hospitalization (immediate postpartum, or IPP-LARC) and instead seek outpatient insertion at a subsequent visit – typically 4 to 8 weeks postpartum, when many LARC insertions occur clinically. Under the 60-day rule, a woman attending a 6-week postpartum visit would have coverage for LARC insertion, but if that visit were delayed even slightly, she could arrive after coverage termination. The 12-month extension provides a substantially longer window for outpatient LARC access, and it also covers the ongoing monitoring visits and potential removal/replacement that LARC requires.

2.3 IPP-LARC Reimbursement Policies

A distinct but related policy lever involves Medicaid reimbursement reforms that unbundle IPP-LARC from the global obstetric fee. This is conceptually different from the coverage extension: IPP-LARC reimbursement addresses a *supply-side billing barrier* at the point of delivery, while the coverage extension addresses a *demand-side eligibility duration* barrier in the outpatient setting. As of October 2023, 45 states and DC had published Medicaid IPP-LARC reimbursement guidance (ACOG, 2023).

Steenland et al. (2021) found that South Carolina’s policy increased monthly IPP-LARC adoption, with effects 2.2 times larger for adolescents. Critically, only 43% of facilities offered IPP-LARC post-policy, highlighting implementation barriers beyond reimbursement. Rodriguez et al. (2024) found a 0.74 percentage-point increase in IPP-LARC use (from a 0.54% baseline – over 100% relative increase) across 15 states using T-MSIS data on 1,378,885 deliveries. This provides the most relevant benchmark for policy-induced LARC effects, though the IPP-LARC reimbursement mechanism is distinct from the coverage extension studied here.

Marthey and colleagues (2024) in *Health Services Research* examined downstream effects of state Medicaid IPP-LARC reforms on maternal mental health using BRFSS data (2014-2019) and found that adoption was associated with 5.7%-11.5% reductions in the probability of reporting poor mental health days among low-income mothers. This finding demonstrates that LARC policy effects can extend beyond contraceptive outcomes to broader maternal well-being.

2.4 Insurance Coverage and Contraceptive Use

The broader literature provides the theoretical and empirical foundation for hypothesizing that coverage extensions will increase LARC uptake. For LARC, with high upfront device costs (\$800-\$1,300 for an IUD) but low marginal costs after insertion, the demand response to insurance is expected to be large at the extensive margin. Carlin, Fertig, and Dowd (2016) found that the ACA’s elimination of contraceptive cost sharing increased \$0 cost-sharing claims for IUDs from 36.6% to 87.6%. Eliason et al. (2022) found that ACA Medicaid expansion was associated with a 7.0 percentage-point increase in postpartum LARC use. However, the ACA expansion operated through expanding initial eligibility rather than extending duration, and the 7.0 percentage-point effect may represent an upper bound for the coverage extension, which targets women who already had coverage at delivery.

Several studies report null effects, providing important context. Pace et al. (2021) found no significant per-capita LARC effect from Medicaid expansion despite absolute increases in prescriptions, suggesting that per-capita effects can be difficult to detect in aggregate data. Law et al. (2016) found no immediate jump in LARC uptake after the ACA contraceptive mandate. Malone, Essex, Frech, and colleagues (2024) demonstrated that continuity of health insurance affects contraceptive method choice, but the relationship between coverage and LARC is mediated by provider capacity, billing complexity, and patient preferences. These findings suggest that coverage alone may be insufficient when supply-side barriers constrain uptake.

2.5 Racial and Ethnic Disparities in Postpartum Contraceptive Access

The equity dimension of postpartum contraceptive policy is central to the policy motivation. Thiel de Bocanegra et al. (2017) documented that among nearly 200,000 California Medicaid births, Black women were less likely to receive postpartum care, any contraception, and highly effective contraception compared to White women. These disparities operate through multiple channels: differential access to prenatal counseling about postpartum contraception, differential rates of postpartum visit attendance, provider implicit bias in contraceptive counseling, and structural barriers including transportation and childcare.

Eliason (2020) used a DiD design to show that Medicaid expansion was associated with 7.01 fewer maternal deaths per 100,000 live births, with significant reductions among non-Hispanic Black women. While this study focused on mortality rather than contraceptive access, it underscores how postpartum coverage policy intersects with structural racism in maternal health. The 12-month extension has the potential either to narrow or to widen racial disparities in postpartum contraceptive access, depending on whether the barriers facing women of color are primarily financial (coverage-responsive) or non-financial (coverage-independent). This heterogeneity motivates the disaggregated analyses in the planned future research agenda.

2.6 Contemporaneous Policy Shocks

The *Dobbs v. Jackson Women’s Health Organization* decision (June 2022) introduced a contemporaneous shock to contraceptive demand. A Becker Friedman Institute working paper (2025) found a 15% increase in LARC insertions in abortion-hostile states immediately after *Dobbs*, though differences were short-lived for most groups, converging by September 2022. If abortion-hostile states were slower to adopt the postpartum extension, the *Dobbs*-induced LARC increase in control states could bias estimates downward. We address this through heterogeneous treatment effect analyses by state abortion policy environment (Section 5).

The most consequential contemporaneous shock for this study is the Medicaid unwinding. The FFCRA continuous enrollment provision expired on March 31, 2023, after which states were permitted to resume eligibility redeterminations and disenroll individuals who no longer met eligibility criteria. CMS estimated that up to 15 million people could lose Medicaid coverage during the redetermination period.

The unwinding is particularly problematic for DiD designs studying 2022-2023 policies because it represents a common shock that differentially affected states based on their pre-unwinding enrollment levels, the pace of their redetermination processes, and the extent to which pandemic-era enrollment gains included individuals who would not have qualified absent the continuous enrollment provision. States that expanded Medicaid under the ACA and adopted the postpartum extension – which substantially overlap – tended to have larger pandemic-era enrollment gains and thus larger unwinding effects. This creates a mechanical correlation between treatment status and the unwinding shock that contaminates raw difference-in-differences comparisons.

The unwinding affects our estimates through at least two channels. First, the shrinking denominator (total Medicaid enrollment) mechanically changes per-capita rates even if the numerator (LARC prescriptions) is unchanged. Second, the composition of the remaining Medicaid population changes as healthier, less medically complex enrollees disenroll first, potentially altering the average utilization intensity. Our extended drug analysis provides direct evidence of this confound by showing that negative control drugs with no postpartum relevance also show negative TWFE coefficients (Section 5.7).

2.7 Econometric Methods for Staggered Adoption

This paper employs modern DiD methods designed for settings with staggered treatment adoption and potentially heterogeneous treatment effects. Goodman-Bacon (2021) showed that the TWFE estimator is a variance-weighted average of all possible 2x2 comparisons in a panel, and that these weights can be negative when treatment effects are heterogeneous or dynamic. In staggered settings, some comparisons use already-treated units as controls, which can produce biased estimates if treatment effects evolve over time. Callaway and Sant’Anna (2021) estimate group-time ATTs that avoid contamination from already-treated comparisons. Gardner (2022) proposes a two-stage procedure (did2s) that estimates fixed effects using untreated observations only. Dube, Girardi, Jorda, and Taylor (2023) develop a local projections approach (LPDID) that provides a flexible, semi-parametric alternative. We implement all three to ensure that our null finding is not an artifact of estimator choice, and we report the Goodman-Bacon decomposition to assess the contribution of potentially problematic comparisons.

2.8 SDUD as a Research Tool

The Medicaid State Drug Utilization Data, maintained by CMS, provides quarterly state-level counts and expenditures for all outpatient drugs paid by Medicaid, identified by NDC. Wen, Hockenberry, Borders, and Druss (2017) demonstrated its utility for tracking state-level prescription drug utilization in response to policy changes. SDUD captures only pharmacy-dispensed drugs and does not capture device-based LARC provision during inpatient delivery stays (IPP-LARC), because devices bundled into DRG payments are excluded by statute and unbundled IPP-LARC billing flows through facility claims rather than the pharmacy point of sale. States perform HCPCS-to-NDC crosswalks inconsistently. This measurement limitation – which our combined panel with T-MSIS outpatient claims is designed to address – is a central feature of the paper’s analytical

framework.

3. Data and Sample Construction

3.1 Data Sources

3.1.1 Medicaid State Drug Utilization Data (SDUD) The outcome data come from the CMS Medicaid State Drug Utilization Data, which reports quarterly state-level counts and expenditures for all outpatient drugs paid by Medicaid, identified by 11-digit National Drug Code (NDC). We downloaded annual SDUD files for 2016-2024 from the CMS Medicaid.gov portal (CMS, 2024). Each record represents a unique state-quarter-NDC-utilization type observation, where utilization type distinguishes fee-for-service (FFSU) and managed care organization (MCOU) claims. We aggregated both to capture the full Medicaid pharmacy benefit.

We identified LARC products using nine NDC codes spanning seven FDA-approved products: four hormonal IUDs (Mirena [2 NDCs], Kyleena, Skyla, Liletta), one copper IUD (Paragard), and two subdermal implants (Nexplanon [2 NDCs, reflecting the 2021 Organon spinoff from Merck], Implanon [discontinued]). We aggregated LARC prescription counts to the state-quarter level.

For the extended drug analysis, we identified NDC codes for eight postpartum-relevant treatment drug categories – short-acting contraceptives, antidepressants (SSRIs/SNRIs), antihypertensives, thyroid medications, diabetes medications, non-benzodiazepine anxiolytics, anticoagulants, and medications for opioid use disorder (MOUD) – and four negative control categories with no clinical connection to postpartum care: statins, BPH medications, gout medications, and Alzheimer’s drugs.

3.1.2 Medicaid Enrollment Data Denominators come from the CMS Medicaid Budget and Expenditure System (MBES), which reports monthly state-level Medicaid enrollment counts. We averaged the three monthly values within each calendar quarter to construct quarterly enrollment denominators. The MBES data do not disaggregate enrollment by age or sex; therefore, we use total Medicaid enrollment as the denominator. This choice is consistent with prior SDUD-based research (Wen et al., 2017) and is conservative, as it attenuates estimated effects by inflating the denominator beyond the postpartum-eligible population.

3.1.3 DOGE T-MSIS Outpatient Claims Data To capture LARC procedures delivered outside the pharmacy channel, we obtained outpatient claims data from the Department of Government Efficiency (DOGE) release of T-MSIS data. These data were made publicly available in February 2026 as part of the DOGE/HHS transparency initiative and report state-quarter-level counts of outpatient procedures by HCPCS and CPT code. We identified LARC procedures billed through HCPCS J-codes (J7296, J7297, J7298, J7300, J7301, J7307) and CPT procedure codes for IUD and implant insertion, removal, and replacement. Like SDUD, the outpatient claims data do not capture inpatient IPP-LARC. The T-MSIS outpatient data are available for 1,357 state-quarter observations with non-missing claims.

Several data quality limitations warrant discussion. First, cell suppression (rows with fewer than 12 claims dropped) could systematically exclude small states with low LARC volumes. Second, the 26% missing rate necessitates zero-imputation for the combined panel; we assess sensitivity by estimating on the T-MSIS subsample (Table C5). Third, the data lack demographic information,

so we cannot restrict to postpartum women. Fourth, as a recent DOGE/HHS release, these data have not been validated against official CMS T-MSIS/TAF research files.

The T-MSIS outpatient claims data complement SDUD by capturing LARC devices dispensed or placed during outpatient clinical visits and billed as medical (rather than pharmacy) benefits. However, like SDUD, the outpatient claims data do not capture LARC devices placed during inpatient delivery stays (IPP-LARC), which are billed through facility claims (UB-04 type 13x).

We constructed three outcome measures: (1) pharmacy-dispensed LARC per 1,000 enrollees (SDUD), (2) outpatient LARC procedures per 1,000 enrollees (T-MSIS), and (3) combined LARC utilization per 1,000 enrollees (summing pharmacy and procedure counts). The combined measure provides the most comprehensive available estimate of outpatient LARC utilization, though it still excludes inpatient IPP-LARC.

3.1.4 National Vital Statistics System (NVSS) The planned reduced-form analysis of interpregnancy intervals requires restricted-use natality data from the National Vital Statistics System (NVSS), maintained by the National Center for Health Statistics (NCHS). The NVSS restricted natality files contain state-level birth records with detailed maternal characteristics including previous birth dates (allowing computation of interpregnancy intervals), maternal age, race/ethnicity, parity, insurance status at delivery, and gestational age. At the time of writing, the restricted NVSS natality files for 2022-2024 have not yet been obtained, and the reduced-form IPI analysis remains a planned future component. The current paper focuses exclusively on the first-stage analysis of LARC and other medication utilization.

3.1.5 State Policy Data Treatment timing comes from the KFF Medicaid Postpartum Coverage Extension Tracker (KFF, 2024), cross-referenced with CMS SPA approval records and the National Academy for State Health Policy tracker. We assigned treatment to the first full calendar quarter following CMS approval. Treatment dates range from 2021 Q3 (Illinois) through 2024 Q3 (Nevada). Five jurisdictions serve as never-treated controls: Arkansas, Florida, New Jersey, Virginia, and Wisconsin.

3.2 Sample Construction

The primary analytic sample is a balanced state-quarter panel comprising 51 jurisdictions (50 states plus the District of Columbia) observed over 36 quarters (2016 Q1 through 2024 Q4), yielding 1,836 observations for the pharmacy channel. Each observation records total LARC prescriptions, prescriptions by device type (IUD vs. implant) and utilization type (FFS vs. MCO), extended drug category counts, total Medicaid enrollment, and treatment status indicators. The outpatient claims panel includes 1,357 state-quarter observations with non-missing procedure counts. The combined panel ($N = 1,836$) sums pharmacy prescriptions and outpatient procedure counts, with procedure counts set to zero for state-quarters without T-MSIS data.

We use a balanced panel rather than an unbalanced panel to avoid compositional changes in the state sample over time. For the combined panel, we assess sensitivity to the zero-imputation decision by estimating models on the T-MSIS subsample only (Table C5). We aggregate across all utilization types (FFS and MCO) for the primary analysis, since the postpartum extension covers women in both delivery systems. We report FFS-only and MCO-only estimates as heterogeneity analyses.

Negative control drugs were selected to satisfy two criteria: (a) no clinical connection to postpartum care and (b) sufficient prescription volume for stable rate estimation. Statins (used for hyperlipidemia), BPH medications (used for benign prostatic hyperplasia, a male-only condition), gout medications, and Alzheimer’s drugs satisfy both criteria. The choice of these specific controls allows us to test whether observed changes in drug utilization rates reflect postpartum-specific policy mechanisms or broader Medicaid enrollment dynamics.

The primary outcome is LARC utilization per 1,000 Medicaid enrollees, measured separately for each channel. For the pharmacy channel, treated states average 0.954 per 1,000 (SD = 0.551) compared with 0.722 per 1,000 (SD = 0.496) in never-treated states. For outpatient procedures, the mean rate is 0.422 per 1,000 (SD = 0.660) among treated states and 0.377 per 1,000 (SD = 0.292) among never-treated states. The combined rate averages 1.242 per 1,000 (SD = 0.751) in treated states and 1.015 per 1,000 (SD = 0.546) in never-treated states (Table C1). IUDs account for approximately 82% of all LARC prescriptions in SDUD, with implants accounting for the remainder. The outpatient claims channel accounts for approximately 16% of total LARC volume among treated states and 27% among never-treated states.

An important feature of the outcome measure is denominator dilution: postpartum-eligible women represent only approximately 1.9% of total Medicaid enrollment, yielding a scaling factor of roughly 57x. This dilution is the primary source of limited statistical power (Section 5.10). The choice is conservative – it cannot produce false positives – but can produce false negatives if the true effect is moderate in magnitude but small relative to total enrollment.

4. Empirical Strategy

4.1 Identification Strategy

The identification strategy exploits staggered adoption of the 12-month extension across states between 2021 Q3 and 2024 Q3. The key identifying assumption is that, absent the extension, treated and comparison states would have followed parallel trends in LARC utilization. This assumption is supported by the event study (Section 5.4). Identifying variation comes from (1) treated vs. never-treated states and (2) early vs. later adopters before the latter are treated. The CS and did2s estimators use both sources of variation.

Several threats to identification warrant discussion. First, adoption is not randomly assigned; states with more generous Medicaid programs may have adopted earlier. We address this through the event study and pre-trends analysis. Second, the compressed adoption window limits variation in treatment timing. Third, the concurrent unwinding and *Dobbs* create common shocks that may differentially affect treated and comparison states. The DDD framework and heterogeneity analyses address these concerns, though imperfectly.

4.2 Two-Way Fixed Effects (TWFE)

Our baseline specification is:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot D_{st} + \varepsilon_{st}$$

where y_{st} is the LARC rate per 1,000 enrollees in state s and quarter t , α_s are state fixed effects, γ_t are quarter fixed effects, D_{st} indicates implementation of the extension, and ε_{st} is the error term.

Standard errors are clustered at the state level (Bertrand, Duflo, and Mullainathan, 2004). The state fixed effects absorb all time-invariant state-level differences; the quarter fixed effects absorb all national-level trends. The coefficient β is identified from within-state changes coinciding with adoption, relative to comparison states.

We implement the Goodman-Bacon (2021) decomposition. Already-treated-as-control comparisons carry 36% of total weight with a mean estimate of -0.041, while clean comparisons are centered near zero, suggesting slight downward bias in the TWFE estimate and motivating the heterogeneity-robust estimators below.

4.3 Two-Stage Difference-in-Differences (did2s)

The Gardner (2022) did2s estimator estimates fixed effects using only untreated observations in the first stage, then regresses the residualized outcome on the treatment indicator in the second stage. This avoids bias from already-treated units serving as controls. Standard errors use the bootstrap procedure recommended by Gardner (2022).

4.4 Local Projections DiD (LPDID)

Following Dube et al. (2023), LPDID estimates the treatment effect at each post-treatment horizon using separate regressions with only not-yet-treated and never-treated controls, avoiding negative weighting while allowing dynamic treatment effects. The LPDID sample is smaller ($N = 1,066$ for pharmacy; $N = 582$ for claims) because the approach requires observations at each horizon relative to treatment.

4.5 Event Study

We estimate an event study specification:

$$y_{st} = \alpha_s + \gamma_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[t - E_s = k] + \varepsilon_{st}$$

where E_s is the first treatment quarter and k indexes quarters relative to treatment onset, with leads and lags from $k = -8$ to $k = +8$. The reference period is $k = -1$. Under parallel trends, pre-treatment leads should be close to zero. We assess pre-period coefficients both visually and through a joint test.

We also estimate the Callaway-Sant’Anna (2021) event study, which avoids TWFE contamination by estimating group-time ATTs using both never-treated and not-yet-treated comparisons. Near-zero pre-treatment CS ATTs provide stronger evidence for parallel trends than the standard TWFE event study.

4.6 Heterogeneity

We estimate TWFE models separately for several outcome subsets to assess whether effects vary by device type, delivery system, or policy environment:

- **Device type:** IUD-only and implant-only outcomes, to test whether the coverage extension differentially affects the two classes of LARC. IUDs and implants have different insertion procedures, cost profiles, and provider training requirements.

- **Delivery system:** FFS-only and MCO-only claims, to test whether effects differ across Medicaid delivery systems. Managed care organizations may have different formulary structures and prior authorization requirements.
- **Abortion policy environment:** Treatment effects stratified by state abortion policy classification to assess whether the *Dobbs* decision confounds estimates differentially.

4.7 Robustness

We conduct a comprehensive robustness battery:

1. **Specification curve:** 14 alternative specifications varying outcome transformations (levels, winsorized, log, inverse hyperbolic sine, extensive margin), sample restrictions (excluding COVID quarters, dropping early or late adopters, large states only, SPA adopters only), weighting (unweighted vs. population-weighted), time trends (state-specific linear trends), and panel length (2018-2024 only).
2. **Randomization inference:** 1,000 permutations of treatment assignment to construct the exact null distribution, providing valid inference even with a small number of clusters.
3. **Leave-one-out jackknife:** 51 re-estimations, each dropping one state, to assess whether any single state drives the result. This is particularly important given the thin control group (5 never-treated states).
4. **Placebo drug test:** Using short-acting contraceptive prescriptions as the outcome to confirm the TWFE design is not generating false positives for pharmacy-dispensed contraceptives.

4.8 Extended Drug Analysis and Triple-Difference

4.8.1 Rationale The extended drug analysis serves two distinct purposes. First, it functions as a *diagnostic* tool: if the postpartum coverage extension is the only relevant policy change, then drugs with no clinical connection to postpartum care should show null TWFE coefficients. Observing significant negative coefficients for negative controls reveals a common shock affecting all drug categories simultaneously. Second, it provides a framework for constructing a *triple-difference* (DDD) estimator that nets out the common confound by differencing a treatment drug against a negative control drug.

4.8.2 DDD Specification For each treatment drug category d , the DDD outcome is:

$$\tilde{y}_{st}^d = \log(1 + y_{st}^d) - \frac{1}{K} \sum_{k=1}^K \log(1 + y_{st}^k)$$

where y_{st}^k is the rate for negative control drug k . The log1p transformation renders drug categories with different absolute scales comparable (statins average 12 per 1,000 while LARC averages 0.98). The coefficient β^{DDD} captures the differential effect net of any common shock. This approach is valid only if the unwinding affects LARC and controls proportionally on the log scale – an assumption we cannot directly test. We verify sensitivity using raw-rate, inverse-hyperbolic-sine, and pre-period-index versions, as well as single-control contrasts.

The current paper is structured as a first-stage analysis within a broader research design. The ultimate policy question is whether postpartum coverage extensions improve birth spacing (reduce short interpregnancy intervals). The causal chain runs from coverage extension to contraceptive access (first stage) to birth spacing (reduced form). The first-stage analysis reported here estimates

the coverage-to-LARC link. The reduced-form analysis – using restricted NVSS natality data to estimate the effect on births within 18 months – is planned but awaits data access. The first-stage results inform whether a sufficiently strong instrument exists for a Wald ratio estimate.

5. Results

5.1 Descriptive Statistics

Table 1 presents summary statistics. Treated states average 1,024 LARC prescriptions per state-quarter (0.95 per 1,000 enrollees), compared with 1,241 prescriptions (0.72 per 1,000) in never-treated states. The higher per-capita rate in treated states despite lower absolute counts reflects smaller average Medicaid enrollment (1.33 million vs. 1.80 million). IUDs dominate, with treated states averaging 930 IUD versus 93 implant prescriptions per quarter. Placebo drug rates are similar across groups (0.55 vs. 0.57 per 1,000), supporting the parallel trends assumption. Table C1 reports combined panel descriptive statistics; outpatient LARC procedures average 0.42 per 1,000 among treated states and 0.38 among never-treated states.

5.2 Main Results: LARC

The postpartum coverage extension has no statistically significant effect on pharmacy-dispensed LARC under any estimator (Table 2). The TWFE estimate is -0.010 per 1,000 enrollees (SE = 0.091, $p = 0.915$, 95% CI: [-0.192, 0.173]). The did2s estimate is -0.030 (SE = 0.102, $p = 0.769$). The LPDID estimate is -0.022 (SE = 0.066). All point estimates are substantively negligible relative to the baseline mean of 0.98.

The consistency across estimators is notable. The Bacon decomposition reveals that already-treated comparisons carry 36% of weight, yet the did2s estimator, which avoids this contamination, produces a nearly identical estimate (-0.030 vs. -0.010). The LPDID also confirms the null (-0.022). This convergence provides strong evidence that the null is not an artifact of estimation approach.

The 95% CI can rule out effects larger than 0.173 per 1,000 total enrollees (~18% of baseline), corresponding to ~9.8 per 1,000 postpartum women. The CI cannot rule out smaller positive effects that might be clinically meaningful.

5.3 Combined Panel Results

Table C2 reports TWFE results across channels. The combined estimate is -0.055 (SE = 0.105, $p = 0.604$), confirming the null when outpatient procedures are included. The procedures-only estimate is -0.119 (SE = 0.079, $p = 0.140$) – the largest in absolute value but still insignificant. Winsorized specifications yield nearly identical results. The did2s confirms the null across channels: combined = -0.076 ($p = 0.495$), pharmacy = -0.030 ($p = 0.769$; Table C3). LPDID estimates are similarly null across all channels (Tables C6a-c).

Channel-specific heterogeneity analysis (Table C4) reveals null effects across all outcomes. Placebo tests pass across all channels: pharmacy = 0.043 ($p = 0.754$), claims = -0.114 ($p = 0.385$), combined = 0.067 ($p = 0.723$). Models estimated on the T-MSIS subsample only (Table C5) produce consistent results (combined: -0.127, $p = 0.248$; pharmacy: -0.012, $p = 0.878$).

Figure C1 displays raw trends by channel showing parallel trajectories. Figure C2 presents channel-specific event studies confirming parallel pre-trends and null post-treatment effects.

5.4 Event Study

Figure 2 displays pharmacy-channel event study coefficients. Pre-treatment coefficients ($k = -8$ through $k = -2$) are centered around zero with no significant departures, confirming parallel trends. The largest pre-treatment coefficient is 0.179 at $k = -6$ (SE = 0.125, $p > 0.10$). Post-treatment coefficients ($k = 0$ through $k = +8$) are slightly positive but imprecise, ranging from 0.030 ($k = 0$) to 0.141 ($k = +6$), with confidence intervals spanning zero at all horizons. The pattern is consistent with a precisely estimated null rather than an underpowered positive effect.

5.5 Heterogeneity

Table 4 reports TWFE estimates by device type and delivery system. Effects are null for IUDs (0.015, SE = 0.061, $p = 0.800$) and implants (-0.025, SE = 0.059, $p = 0.670$). The FFS-only estimate is -0.111 (SE = 0.070, $p = 0.117$) and the MCO-only estimate is 0.102 (SE = 0.078, $p = 0.200$), suggesting a possible offsetting pattern, though neither is significant. The placebo drug test yields 0.043 ($p = 0.754$), confirming no false positive.

5.6 Robustness

The specification curve (Figure 4; Table 6) confirms the null across all 14 specifications. Coefficients range from -0.070 (dropping late adopters) to 0.037 (log transformation), with none significant. Randomization inference yields $p = 0.917$, indicating the observed coefficient is well within the null distribution. Leave-one-out analysis shows no single state drives the result (coefficient range: [-0.055, 0.034]).

5.7 Extended Drug Analysis

Table 7 reports TWFE estimates for all 12 drug categories. A striking pattern emerges: every drug category except Alzheimer’s drugs shows a negative TWFE coefficient, and most are statistically significant. Critically, the negative control drugs – which have no clinical connection to postpartum care – exhibit substantial negative effects: statins (-12.0 per 1,000, $p < 0.001$), gout medications (-0.67 per 1,000, $p = 0.024$), and BPH drugs (-0.80 per 1,000, $p = 0.044$). The only exception is Alzheimer’s drugs (+0.13, $p = 0.159$).

This pattern is diagnostic of a confound that depresses all drug utilization rates among the treated group relative to controls during the post-treatment period. The most likely explanation is the Medicaid unwinding: the end of the continuous enrollment provision in March 2023 triggered mass disenrollment that disproportionately affected states that had already adopted the postpartum extension (which tended to be expansion states with larger pandemic-era enrollment gains). As healthier enrollees disenrolled, remaining beneficiaries may have had higher baseline utilization, but the shrinking denominator interacted with compositional changes to produce apparent declines in per-capita rates. The did2s estimates (Table 8) corroborate these findings across all drug categories.

5.8 Scale-Adjusted Triple-Difference Results

Table 9 reports the re-estimated DDDs on a relative scale, differencing each treatment drug category against the pooled negative control drugs after log1p transformation. Under this more defensible scaling, the earlier pattern of large positive DDD effects does not persist. LARC has the largest positive estimate (+0.094, SE = 0.055, $p = 0.095$), but short-acting contraceptives (+0.031, $p = 0.509$), anticoagulants (+0.099, $p = 0.250$), antidepressants (+0.018, $p = 0.473$), and the remaining

categories are all imprecise. Figure 3 presents these scale-adjusted DDD estimates for all eight treatment drug categories.

This result changes the interpretation of the extended drug analysis. The raw TWFE pattern remains useful as a diagnostic of the unwinding confound, because even negative control drugs show negative post-treatment estimates. But once the DDD is placed on a relative scale comparable across drug classes, the evidence no longer supports strong conclusions about multiple medication categories responding positively to the postpartum extension.

5.9 LARC-Specific Triple-Difference Sensitivity

The LARC-specific DDD remains directionally positive but depends materially on scale and control set. Using actual negative controls on the primary log1p scale: TWFE DDD = +0.094 (SE = 0.055, $p = 0.095$, 95% CI [-0.017, 0.204]); did2s = +0.109 (SE = 0.062, $p = 0.078$, 95% CI [-0.012, 0.231]). Inverse-hyperbolic-sine and pre-period-index normalizations attenuate to +0.105 ($p = 0.135$) and +0.079 ($p = 0.423$). Individual-control log1p contrasts are mixed: statin comparison (+0.187, $p = 0.005$), gout (+0.108, $p = 0.070$), BPH (+0.094, $p = 0.114$), Alzheimer’s (-0.015, $p = 0.793$). The DDD evidence is directionally positive but should be interpreted as exploratory rather than definitive.

5.10 Power Analysis and Minimum Detectable Effects

Formal power analysis (Table 11) quantifies why the raw null is partially uninformative. At 80% power and $\alpha = 0.05$, the MDE for the pharmacy channel is 0.255 per 1,000 total enrollees (26% of baseline). For the combined channel, the MDE is 0.293 (23% of baseline). The pharmacy MDE exceeds the Rodriguez et al. (2024) benchmark of 0.185, meaning the design cannot detect effects of that magnitude. The Eliason et al. (2022) benchmark of 1.75 is well within the detectable range. The design can rule out very large effects but cannot distinguish between a true zero and a small-to-moderate positive effect.

5.11 Callaway-Sant’Anna Event Study

Figure 5 presents the CS event study. Pre-treatment ATTs are near zero: $t = -4$ through $t = -1$ coefficients are all within 0.02 of zero, supporting parallel trends. Post-treatment ATTs are consistently positive, ranging from +0.038 ($t = 0$) to +0.128 ($t = +6$), with an overall weighted post-treatment ATT of +0.052 (SE = 0.028). Most individual post-treatment coefficients are imprecise, reflecting denominator dilution. The consistently positive direction is compatible with a modest positive outpatient effect but insufficient to validate a large causal claim.

6. Discussion

6.1 Interpreting the LARC Results: Informative Null Plus Suggestive Relative Evidence

The LARC analysis is anchored by null outpatient estimates across every channel. Standard TWFE estimates show no detectable effect for pharmacy prescriptions (-0.010, $p = 0.915$), outpatient procedures (-0.119, $p = 0.140$), or the combined measure (-0.055, $p = 0.604$). These nulls are reinforced by did2s, LPDID, the claims-only subsample, the specification curve, randomization

inference, and leave-one-out analysis. The DOGE claims upgrade improves outpatient measurement but does not overturn the reduced-form null.

At the same time, this paper cannot be read as establishing a clean “true zero.” The negative-control analysis shows the unwinding distorted raw post-2022 comparisons, and the scale-adjusted DDDs are directionally positive. The most credible reading: the outpatient first stage is null in standard aggregate models, but those models are contaminated by a broad enrollment shock and underpowered for small-to-moderate effects. Once that shock is addressed on a relative scale, the residual LARC signal is modestly positive but too sensitive to scaling and control choice for a definitive headline.

6.2 The Medicaid Unwinding as a Confound

Our extended drug analysis reveals a finding with broad implications: the unwinding acts as a pervasive confound in contemporaneous DiD designs. Negative control drugs with no postpartum relevance all show negative TWFE coefficients, pointing to a common enrollment and composition shock. This is the paper’s strongest contribution. Any Medicaid policy evaluation spanning 2022-2024 should treat the unwinding as a first-order identification threat. Researchers evaluating other contemporaneous policies – managed care transitions, prior authorization reforms, supplemental payment restructuring – should adopt similar diagnostic strategies using negative control outcomes.

The DDD approach offers one methodological strategy for addressing this confound. However, the LARC results also demonstrate that DDD conclusions are sensitive to how outcomes are scaled and which controls are used, suggesting that the diagnostic value of the approach may exceed its corrective value.

6.3 Pharmacy Versus Clinical Channels

An important interpretation of the null concerns the distinction between channels for LARC provision. LARC devices can be dispensed through at least three Medicaid channels: (1) pharmacy dispensing, billed through the pharmacy benefit (captured by SDUD); (2) outpatient clinical procedures, billed through outpatient medical claims (captured by T-MSIS); and (3) inpatient facility procedures during delivery (not captured by either data source).

The multi-channel analysis reveals that the null is not specific to the pharmacy channel. The outpatient procedure estimate (-0.119, $p = 0.140$) is the largest in absolute value and directionally negative, inconsistent with a positive effect being masked by pharmacy-specific measurement issues. The combined estimate (-0.055, $p = 0.604$) confirms the null even when both outpatient channels are aggregated. This suggests the null is not simply a missing-channel artifact for the outpatient setting. However, the missing inpatient IPP-LARC channel remains a critical gap: if the extension primarily encourages LARC insertion during delivery (by assuring women and providers that the device will be covered and follow-up available), the policy effect could operate entirely through the inpatient channel we cannot observe.

The FFS-MCO heterogeneity provides additional texture. The FFS-only estimate is -0.111 ($p = 0.117$) while the MCO-only estimate is +0.102 ($p = 0.200$), suggesting a possible offsetting pattern between delivery systems. One interpretation is that the extension had a modest positive effect through managed care but a countervailing effect through fee-for-service, possibly reflecting differential administrative responses. However, neither estimate is significant and the difference is not formally tested, so this interpretation remains speculative.

6.4 Comparison with Prior Literature

Our findings remain compatible with Swartz et al. (2025), who found increased postpartum contraceptive utilization in North Carolina after the extension, but they no longer provide equally strong national confirmation. Several factors reconcile the results. Swartz et al. used individual-level Medicaid claims and could identify postpartum beneficiaries directly, avoiding the denominator dilution that attenuates our state-level estimates. Their outcome included all contraceptive utilization, including inpatient LARC and short-acting methods, whereas our directly observed outcomes are outpatient channels only. Finally, the single-state pre-post design cannot separate the extension from broader secular changes, including the unwinding shock we document here.

The comparison with Rodriguez et al. (2024) is particularly instructive. Their study of IPP-LARC reimbursement policies found a 0.74 percentage-point increase from a 0.54% baseline, a large relative effect that operated through a supply-side mechanism (unbundling reimbursement at the point of delivery). Our study examines a demand-side mechanism (extending eligibility duration), and the null outpatient finding is consistent with the hypothesis that demand-side interventions have weaker effects on LARC uptake than supply-side interventions. The IPP-LARC literature consistently shows that reimbursement reform alone is insufficient without facility-level implementation – only 43% of facilities offered IPP-LARC after South Carolina’s policy change (Steenland et al., 2021) – and the coverage extension faces analogous implementation challenges in the outpatient setting.

The Eliason et al. (2022) finding of a 7.0 percentage-point increase from ACA Medicaid expansion operated through expanding initial eligibility rather than extending duration. The ACA expansion brought women into coverage who had no prior Medicaid access, generating a large extensive-margin effect. The 12-month extension targets women who already have coverage at delivery, and the marginal effect of extended duration may be smaller, particularly if most LARC decisions are made during or shortly after delivery, when coverage is already in place.

The broader null is consistent with earlier studies showing that coverage expansions do not mechanically translate into large population-level LARC increases when supply-side barriers persist. Pace et al. (2021) found no significant per-capita effect despite absolute increases, and Law et al. (2016) found no immediate jump after the ACA mandate. Taken together, the literature and the current paper suggest that postpartum coverage may help at the margin, but provider capacity, billing systems, and inpatient delivery-site practices likely remain important constraints on total LARC uptake.

6.5 Limitations

Several limitations warrant discussion.

First, even with the combined panel, our analysis does not capture inpatient IPP-LARC – devices placed during delivery hospitalizations – which may be a dominant channel for postpartum LARC provision. Our null result cannot be interpreted as evidence that the postpartum extension has no effect on total LARC uptake. Studies using T-MSIS/TAF inpatient facility claims are needed to capture this channel.

Second, our denominator is total Medicaid enrollment rather than postpartum-eligible women. This attenuates the estimated effect because the denominator is far larger than the affected population. Postpartum-eligible women represent approximately 1.9% of total enrollment (scaling factor $\sim 57x$). The denominator-adjusted TWFE coefficient is -1.91 per 1,000 postpartum women ($p = 0.74$, 95% CI [-13.5, 9.7]), with wide confidence intervals that confirm limited power. We adopt total enroll-

ment because MBES data do not disaggregate by age or sex, consistent with prior SDUD-based research.

Third, the control group is thin. Only five states serve as never-treated controls, and three (Florida, New Jersey, Virginia) have pending Section 1115 waivers. States with pending waivers may exhibit anticipation effects – providers may begin offering IPP-LARC in expectation of forthcoming coverage – which would bias estimates toward zero. The CS framework mitigates reliance on these states at shorter event horizons by also using not-yet-treated states as controls. The randomization inference p-value of 0.917 addresses the quantity concern but not the quality concern about anticipation.

Fourth, the DOGE T-MSIS data have several quality limitations. Cell suppression (dropping rows with fewer than 12 claims) could bias estimates by excluding small states. The 26% missing rate, combined with zero-imputation, could bias toward the null. The data lack demographic information and have not been validated against official CMS T-MSIS/TAF research files.

Fifth, the study period overlaps with the *Dobbs* decision. We estimated heterogeneous effects by state abortion policy environment: hostile (-0.058 , $p = 0.67$), supportive ($+0.067$, $p = 0.51$), and middle ($+0.073$, $p = 0.67$). A post-*Dobbs* $\times$ hostile-state interaction term is suggestive but not significant ($+0.141$, $p = 0.21$). These results provide no evidence of differential *Dobbs* confounding, though the interaction estimate is imprecise.

Sixth, the compressed adoption window limits heterogeneous treatment effect power across adoption cohorts.

Seventh, the LARC outcome is subject to a compositional break in implant identification. Nexplanon received a new NDC labeler code in 2021 Q3 following Organon’s spinoff from Merck. The SDUD data show zero implant prescriptions under Nexplanon NDCs prior to 2021 Q3, with a sharp discontinuity to positive counts (Figure 6). This pattern is consistent with a labeler code transition rather than a true absence. The break could affect the LARC outcome level, though it would not bias the DiD estimate unless interacting differentially with treatment timing. DDD results are less susceptible because the break is specific to LARC.

6.6 Policy Implications

Our findings carry several implications for policy, practice, and research methodology.

For Medicaid Policy Evaluation Methodology The identification of the Medicaid unwinding as a pervasive confound is perhaps the most broadly relevant finding. Any DiD evaluation of Medicaid policies adopted during 2022-2024 must contend with the unwinding as a simultaneous shock. The fact that negative control drugs with no postpartum relevance all show significant negative TWFE coefficients is a striking empirical demonstration of this problem. Researchers evaluating other contemporaneous policies – including managed care transitions, prior authorization reforms, and supplemental payment restructuring – should adopt similar diagnostic strategies using negative control outcomes. Our DDD approach offers one methodological strategy, but the LARC results demonstrate that DDD conclusions are sensitive to scaling and control choice, suggesting that the diagnostic value may exceed the corrective value.

For Postpartum Coverage Policy For policymakers, the implication is not that the extension is ineffective. Rather, the paper shows that an apparent null can coexist with meaningful

confounding and limited power. The scale-adjusted LARC DDDs are consistent with a modest positive outpatient response, but they are not precise enough to stand alone. The denominator dilution problem means that even a clinically meaningful effect on the target population would be extremely difficult to detect in aggregate data. Effects may also operate through unobserved channels (inpatient LARC), on unmeasured outcomes (mental health, substance use treatment, chronic disease management), or on populations too small relative to total enrollment to detect. The North Carolina evidence from Swartz et al. (2025) is arguably more policy-relevant for the question of whether the extension helps postpartum women, even though it lacks the causal design of our national analysis.

For Clinical Practice The null outpatient finding, combined with the strong effects of IPP-LARC reimbursement policies documented by Rodriguez et al. (2024) and Steenland et al. (2019, 2021), reinforces the importance of inpatient delivery-site LARC provision. Counseling and offering LARC at the delivery hospitalization remains the most proximate intervention. The null is consistent with the hypothesis that most women who want LARC have already received it by the time they would access outpatient insertion in months 2-12 postpartum, or that supply-side barriers in outpatient settings prevent the coverage extension from translating into utilization. Additional effects may exist in the unmeasured inpatient IPP-LARC channel.

6.7 Future Research

Key next steps include: (1) T-MSIS/TAF inpatient analysis of IPP-LARC insertions, which is the remaining unmeasured channel; (2) reduced-form analysis using restricted NVSS natality data for effects on short IPI; (3) Wald ratio estimation combining first-stage and reduced-form estimates; and (4) racial/ethnic heterogeneity analysis to assess whether the extension narrows or widens disparities in contraceptive access.

7. Conclusion

This paper develops a multi-data-source framework for evaluating postpartum Medicaid coverage extensions and identifies the Medicaid unwinding as a dominant confound. Using SDUD and DOGE T-MSIS outpatient claims across 51 jurisdictions over 36 quarters, we document three principal findings.

First, the DOGE claims upgrade improves outpatient measurement but does not overturn the core null: pharmacy, outpatient-procedure, and combined estimates are all close to zero and imprecise. The null is robust across three estimators (TWFE, did2s, LPDID), 14 alternative specifications, randomization inference, leave-one-out sensitivity, and placebo tests. The CS event study confirms flat pre-trends and a consistently positive but imprecise post-treatment path. Heterogeneity analyses reveal null effects across device types, delivery systems, data channels, and abortion policy environments.

Second, the Medicaid unwinding depresses many drug utilization categories – including negative controls with no postpartum relevance (statins: -12.0 per 1,000, $p < 0.001$; BPH: -0.80, $p = 0.044$) – making raw post-2022 comparisons unreliable. This is the paper’s most portable methodological contribution: any Medicaid policy evaluation during 2022-2024 should test for unwinding contamination using negative control diagnostics.

Third, scale-adjusted LARC DDDs are suggestive rather than definitive. The log1p DDD is positive under TWFE (+0.094, $p = 0.095$) and did2s (+0.109, $p = 0.078$), but alternative normalizations attenuate further and individual-control contrasts are mixed. The most credible conclusion is that postpartum extensions may modestly increase outpatient LARC utilization relative to secular trends, but the current data do not pin that effect down sharply enough for a strong first-stage estimate.

These findings contribute to the growing literature on insurance coverage expansions and LARC uptake by showing that the relationship between coverage duration and contraceptive utilization is more nuanced than simple demand-side models predict. The null outpatient result is consistent with a supply-constrained model in which LARC provision depends primarily on inpatient delivery-site practices and provider-level implementation, with coverage duration playing a secondary role. The findings also contribute to the Medicaid policy evaluation methodology literature by demonstrating why contemporaneous evaluations can be misleading during the unwinding and by providing a concrete diagnostic framework for identifying and partially addressing this confound.

For the broader research agenda on postpartum coverage and birth spacing, this paper establishes the outpatient evidence base and identifies the key measurement priorities for future work. Stronger first-stage evidence likely requires inpatient IPP-LARC claims or postpartum-specific denominators, both of which are available through T-MSIS/TAF restricted research files. Evaluating downstream effects on interpregnancy intervals requires restricted NVSS natality data. The multi-data-source methodology and the confound-diagnostic framework developed here remain useful even though the paper’s strongest substantive claim is more modest than initial expectations: the coverage extension does not generate a large, precisely estimated increase in outpatient LARC utilization, but the combination of confounding, limited power, and missing channels means that a genuinely important positive effect cannot be ruled out. Resolving this ambiguity is the central task for the next phase of research.

The policy case for 12-month postpartum Medicaid coverage extensions does not depend solely on LARC outcomes. The extension covers a broad range of postpartum services – mental health treatment, substance use disorder care, chronic disease management, and preventive care – for which the evidence base is still being developed. Even if LARC effects prove modest, the broader maternal health benefits of extended coverage may justify the policy on their own terms. This paper provides one carefully bounded piece of the overall evidence picture: on the specific question of outpatient LARC utilization, the extension’s effects are either small, concentrated in unmeasured channels, or obscured by the concurrent enrollment shock. Future work will determine which of these interpretations holds.

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Tables

Table 1. Descriptive Statistics

Variable	Group	N	Mean	SD	Min	Median	Max
LARC prescriptions (count)	Treated	1,323	1,024	1,145	0	626	8,251
	Never-treated	180	1,241	890	0	1,143	3,296
LARC per 1,000 enrollees	Treated	1,323	0.954	0.551	0.000	0.861	5.831
	Never-treated	180	0.722	0.496	0.000	0.549	1.739
IUD prescriptions	Treated	1,323	930	992	0	569	7,152
	Never-treated	180	1,007	662	0	1,094	2,327
Implant prescriptions	Treated	1,323	93	345	0	0	3,954
	Never-treated	180	233	397	0	0	1,402
Total Medicaid enrollment	Treated	1,323	1,327,667	1,847,956	51,410	806,679	13,093,534
	Never-treated	180	1,796,200	1,142,406	743,012	1,381,586	4,897,219
Placebo drugs per 1,000	Treated	1,323	0.550	0.661	0.000	0.333	3.123
	Never-treated	180	0.571	0.819	0.000	0.254	3.348

Notes: Balanced panel of 51 jurisdictions observed quarterly from 2016 Q1 through 2024 Q4 (N = 1,836). Treated states are those that adopted the 12-month postpartum Medicaid coverage extension during the study period (46 states + DC). Never-treated states: AR, FL, NJ, VA, WI.

LARC = long-acting reversible contraceptive; IUD = intrauterine device. SDUD data from CMS; enrollment data from MBES.

Table 2. Main Results: Effect of Postpartum Extension on LARC Prescriptions per 1,000 Enrollees

Specification	Coefficient	SE	p-value	95% CI	N
TWFE	-0.010	0.091	0.915	[-0.192, 0.173]	1,836
TWFE (winsorized 99th)	-0.002	0.090	0.984	[-0.183, 0.179]	1,836
did2s (Gardner 2022)	-0.030	0.102	0.769	[-0.234, 0.174]	1,836
LPDID (Dube et al. 2023)	-0.022	0.066	0.739	[-0.155, 0.111]	1,066

Notes: All models include state and quarter fixed effects. Standard errors clustered at the state level. TWFE = two-way fixed effects; did2s = Gardner (2022) two-stage difference-in-differences; LPDID = local projections difference-in-differences. The outcome is LARC prescriptions per 1,000 total Medicaid enrollees.

Table 3. Event Study Coefficients

Relative Quarter (k)	Coefficient	SE	95% CI
-8	-0.081	0.132	[-0.346, 0.185]
-7	0.019	0.100	[-0.182, 0.220]
-6	0.179	0.125	[-0.072, 0.430]
-5	0.022	0.077	[-0.132, 0.176]
-4	0.013	0.077	[-0.143, 0.169]
-3	0.024	0.061	[-0.098, 0.146]
-2	0.026	0.037	[-0.048, 0.099]
-1	0.000	—	[ref.]
0	0.030	0.042	[-0.055, 0.116]
+1	0.047	0.069	[-0.092, 0.185]
+2	0.073	0.092	[-0.112, 0.259]
+3	0.071	0.163	[-0.259, 0.400]
+4	0.090	0.183	[-0.279, 0.457]
+5	0.084	0.204	[-0.328, 0.495]
+6	0.141	0.229	[-0.320, 0.602]
+7	0.063	0.254	[-0.449, 0.575]
+8	0.118	0.279	[-0.443, 0.679]

Notes: Event study estimates from TWFE specification with leads and lags. Reference period is $k = -1$ (one quarter before treatment). Standard errors clustered at the state level.

Table 4. Heterogeneity by Device Type and Delivery System

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
All LARC	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
IUD only	0.015	0.061	0.800	[-0.107, 0.137]	1,836	0.806
Implant only	-0.025	0.059	0.670	[-0.143, 0.093]	1,836	0.173
FFS only	-0.111	0.070	0.117	[-0.252, 0.029]	1,836	0.372
MCO only	0.102	0.078	0.200	[-0.055, 0.258]	1,836	0.608
Placebo drugs (falsif.)	0.043	0.136	0.754	[-0.230, 0.315]	1,836	0.605

Notes: TWFE estimates with state and quarter fixed effects. Standard errors clustered at the state level. FFS = fee-for-service; MCO = managed care organization. Placebo drugs include short-acting contraceptives (Depo-Provera, NuvaRing, selected OCPs).

Note on the dual role of short-acting contraceptives: In Table 4, short-acting contraceptives serve as a falsification test for LARC specifically. In the extended drug analysis (Table 5), they are treated as a treatment-relevant outcome through a broader coverage channel. The scale-adjusted DDD estimate is small and imprecise (+0.031, $p = 0.509$). These roles are complementary: the placebo test speaks to LARC-specific spillovers, while the DDD asks whether broader coverage changes pharmacy-dispensed contraceptive use net of the unwinding.

Table 5. Extended Drug Category Results: TWFE and Triple-Difference

Drug Category	Group	TWFE Coef.	TWFE p	Scale-adjusted DDD Coef.	DDD p
Short-acting contraceptives	Treatment	-1.60	0.074	+0.031	0.509
Antidepressants (SS-RIs/SNRIs)	Treatment	-14.19	0.023	+0.018	0.473
Antihypertensives	Treatment	-25.39	0.002	-0.063	0.130
Thyroid medications	Treatment	-4.72	0.005	-0.025	0.288

Drug Category	Group	TWFE Coef.	TWFE p	Scale-adjusted DDD Coef.	DDD p
Diabetes medications	Treatment	-6.73	0.009	-0.033	0.386
Non-benzo anxiolytics	Treatment	-5.26	0.230	+0.034	0.245
Anticoagulants	Treatment	-0.59	0.569	+0.099	0.250
MOUD (buprenorphine/naloxone)	Treatment	-3.10	0.254	-0.119	0.203
Statins	Neg. control	-11.99	<0.001	—	—
BPH medications	Neg. control	-0.80	0.044	—	—
Gout medications	Neg. control	-0.67	0.024	—	—
Alzheimer’s drugs	Neg. control	+0.13	0.159	—	—

Notes: TWFE = two-way fixed effects with state and quarter fixed effects; DDD = triple-difference, differencing each treatment drug against the pooled negative control drugs after a log1p transformation. TWFE coefficients are expressed as prescriptions per 1,000 Medicaid enrollees; DDD coefficients are relative-scale differences in transformed rates. Standard errors clustered at the state level. N = 1,836 for all specifications.

Table 6. Specification Curve: LARC Results Across 14 Specifications

Specification	Coefficient	SE	p-value	N
Baseline TWFE	-0.010	0.091	0.915	1,836
Winsorized (95th pct)	0.031	0.086	0.721	1,836
Winsorized (99th pct)	-0.002	0.090	0.984	1,836
Log(LARC + 0.001)	0.037	0.291	0.900	1,836
Arcsinh(LARC)	0.013	0.061	0.828	1,836
Extensive margin	-0.000	0.039	0.995	1,836
Excl. COVID quarters	0.004	0.093	0.968	1,581
Population-weighted	-0.040	0.067	0.559	1,836
Drop IL (early adopter)	-0.005	0.095	0.962	1,800
Drop late adopters (2024+)	-0.070	0.085	0.413	1,512
Large states only	-0.033	0.110	0.764	900
SPA adopters only	-0.022	0.095	0.814	1,764
State-specific trends	0.029	0.081	0.721	1,574
2018-2024 only	-0.012	0.080	0.878	1,428

Notes: All models include state and quarter fixed effects. Standard errors clustered at the state level. No specification yields statistical significance at the 0.05 level.

Figures

Figure 1. Raw Trends in LARC Prescriptions per 1,000 Enrollees, Treated vs. Never-Treated States, 2016-2024

[Figure 1]

Notes: Solid line = mean LARC rate in states that adopted the 12-month postpartum extension. Dashed line = mean LARC rate in never-treated states (AR, FL, NJ, VA, WI). Vertical dashed line marks the first treatment quarter (2021 Q3, Illinois).

Figure 2. Event Study: LARC Prescriptions per 1,000 Enrollees

[Figure 2]

Notes: Point estimates and 95% confidence intervals from TWFE event study specification. Reference period is $k = -1$. Pre-treatment coefficients ($k = -8$ to -2) are centered around zero, confirming parallel trends. Post-treatment coefficients are positive but imprecise, with all confidence intervals spanning zero. Standard errors clustered at the state level.

Figure 3. Triple-Difference Forest Plot: Effects by Drug Category

[Figure 3]

Notes: Scale-adjusted DDD point estimates and 95% confidence intervals for each treatment drug category, differenced against pooled negative controls after a $\log_{1p}$ transformation. Dashed vertical line at zero.

Figure 4. Specification Curve: LARC TWFE Coefficients Across 14 Specifications

[Figure 4]

Notes: Each point represents the TWFE coefficient from one of 14 alternative specifications. Whiskers show 95% confidence intervals. All coefficients are centered near zero and none are statistically significant.

Appendix

Appendix Figure A1. Cohort-Specific Trends

[Appendix Figure A1]

Appendix Figure A2. Multi-Drug TWFE Forest Plot

[Appendix Figure A2]

Appendix Figure A3. Drug Category Raw Trends

[Appendix Figure A3]

Appendix Figure A4. Bacon Decomposition (Histogram)

[Appendix Figure A4]

Appendix Figure A5. Bacon Decomposition Scatter

[Appendix Figure A5]

Notes: Weight-by-estimate scatter plot of 1,900 pairwise 2x2 DD estimates from the Goodman-Bacon decomposition, color-coded by comparison type. Already-treated-as-control comparisons carry 36% of total weight with mean estimate -0.041.

Appendix Figure A6. Treatment Timing Distribution

[Appendix Figure A6]

Notes: Bar chart showing the number of states adopting the extension in each quarter, with a cumulative adoption curve. Five jurisdictions remain never-treated.

Appendix Figure A7. Randomization Inference Distribution

[Appendix Figure A7]

Appendix Figure A8. Leave-One-Out Sensitivity

[Appendix Figure A8]

Appendix Table A1. Leave-One-Out Jackknife Results (Selected States)

Dropped State	Coefficient	SE	p-value
Utah	-0.055	0.080	0.500
Wisconsin	0.034	0.085	0.689
Tennessee	-0.027	0.091	0.772
Alaska	-0.026	0.092	0.779
Florida	-0.021	0.095	0.826
Nevada	-0.036	0.091	0.696

Notes: Full results for all 51 jurisdictions available on request. Range of coefficients: [-0.055, 0.034]. No single state drives the null result.

Appendix Table A2. did2s Results for Extended Drug Categories

Drug Category	Coefficient	SE	p-value
Short-acting contraceptives	-1.58	1.084	0.145
Antidepressants	-16.13	8.218	0.050
Antihypertensives	-26.93	9.961	0.007
Thyroid medications	-5.80	2.030	0.004
Diabetes medications	-7.99	3.043	0.009
Non-benzo anxiolytics	-5.06	6.165	0.412
Anticoagulants	-0.74	1.363	0.585
MOUD	-4.31	2.677	0.108
Statins (neg. control)	-13.87	3.963	0.001
Alzheimer's (neg. control)	0.13	0.080	0.109
BPH (neg. control)	-1.07	0.445	0.016
Gout (neg. control)	-0.84	0.342	0.015

Notes: did2s (Gardner, 2022) estimates. Standard errors clustered at the state level. N = 1,836 for all categories.

Appendix C: Combined Panel Analysis (SDUD + T-MSIS Outpatient Claims)

Table C1. Descriptive Statistics: Combined Panel

Variable	Group	N	Mean	SD	Min	Median	Max
LARC per 1,000 (phar- macy)	Treated	1,323	0.954	0.551	0.000	0.861	5.831
	Never- treated	180	0.722	0.496	0.000	0.549	1.739
LARC per 1,000 (outpa- tient claims)	Treated	903	0.422	0.660	0.000	0.159	4.495
	Never- treated	140	0.377	0.292	0.000	0.253	1.264
LARC per 1,000 (com- bined)	Treated	1,323	1.242	0.751	0.000	1.089	6.689
	Never- treated	180	1.015	0.546	0.000	0.964	2.028
LARC Rx count (phar- macy)	Treated	1,323	1,024	1,145	0	626	8,251

Variable	Group	N	Mean	SD	Min	Median	Max
LARC claims (outpatient)	Never-treated	180	1,241	890	0	1,143	3,296
	Treated	1,323	597	2,693	0	13	35,641
LARC volume (combined)	Never-treated	180	386	307	0	392	1,418
	Treated	1,323	1,620	3,288	0	724	40,061
Total Medicaid enrollment	Never-treated	180	1,627	985	0	1,402	4,714
	Treated	1,323	1,327,667	1,847,956	51,410	806,679	13,093,534
Claims channel share	Never-treated	180	1,796,200	1,142,406	743,012	1,381,586	4,897,219
	Treated	1,320	0.161	0.243	0.000	0.022	1.000
Placebo per 1,000 (pharmacy)	Never-treated	172	0.274	0.273	0.000	0.197	1.000
	Treated	1,323	0.550	0.661	0.000	0.333	3.123
Placebo per 1,000 (claims)	Never-treated	180	0.571	0.819	0.000	0.254	3.348
	Treated	903	1.433	1.595	0.000	1.105	14.830
Placebo per 1,000 (combined)	Never-treated	140	1.840	1.357	0.015	1.416	5.237
	Treated	1,323	1.529	1.777	0.000	1.117	15.519
	Never-treated	180	2.002	2.130	0.000	1.162	7.762

Notes: Balanced panel of 51 jurisdictions observed quarterly. Outpatient claims data available for a subset of state-quarters (N = 1,357 with non-missing data). Combined measure sums pharmacy

and outpatient procedure counts. Claims channel share = outpatient claims / combined volume.

Table C2. TWFE Results by Channel

Channel	Coefficient	SE	p-value	95% CI	N	Mean
Combined (pharmacy + procedures)	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278
Combined (win- sorized)	-0.045	0.103	0.663	[-0.253, 0.162]	1,836	1.273
Outpatient procedures only	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403
Outpatient procedures (win- sorized)	-0.108	0.074	0.151	[-0.256, 0.041]	1,357	0.398
Pharmacy dispensing only	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy dispensing (win- sorized)	-0.002	0.090	0.984	[-0.183, 0.179]	1,836	0.975

Notes: All models include state and quarter fixed effects. Standard errors clustered at the state level. Outcomes expressed as LARC utilization per 1,000 total Medicaid enrollees.

Table C3. did2s Results by Channel

Channel	Coefficient	SE	p-value	95% CI	N
Combined (pharmacy + procedures)	-0.076	0.111	0.495	[-0.293, 0.142]	1,836
Pharmacy dispensing only	-0.030	0.102	0.769	[-0.229, 0.169]	1,836

Notes: Gardner (2022) two-stage difference-in-differences. Standard errors clustered at the state level.

Table C4. Heterogeneity: Combined Panel

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
Combined:	-0.055	0.105	0.604	[-0.265,	1,836	1.278
All LARC				0.156]		
Pharmacy:	-0.010	0.091	0.915	[-0.192,	1,836	0.980
All LARC				0.173]		
Pharmacy:	0.015	0.061	0.800	[-0.107,	1,836	0.806
IUD				0.137]		
Pharmacy:	-0.025	0.059	0.670	[-0.143,	1,836	0.173
Implant				0.093]		
Pharmacy:	-0.111	0.070	0.117	[-0.252,	1,836	0.372
Fee-for-				0.029]		
service						
Pharmacy:	0.102	0.078	0.200	[-0.055,	1,836	0.608
Managed				0.258]		
care						
Claims: All	-0.119	0.079	0.140	[-0.277,	1,357	0.403
LARC				0.040]		
procedures						
Placebo:	0.043	0.136	0.754	[-0.230,	1,836	0.605
Pharmacy				0.315]		
Placebo:	-0.114	0.130	0.385	[-0.376,	1,357	1.447
Claims				0.148]		
Placebo:	0.067	0.188	0.723	[-0.310,	1,836	1.674
Combined				0.444]		

Notes: TWFE estimates with state and quarter fixed effects. Standard errors clustered at the state level. Placebo tests confirm no false positives in any channel.

Table C5. Claims-Only Subsample Analysis (T-MSIS States Only)

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Claims:	-0.119	0.079	0.140	[-0.277,	1,357	51	0.403
LARC				0.040]			
proce-							
dures							
Combined:	-0.127	0.108	0.248	[-0.344,	1,428	51	1.424
pharmacy				0.091]			
+ proce-							
dures							

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Pharmacy only (T-MSIS subsample)	-0.012	0.080	0.878	[-0.173, 0.149]	1,428	51	1.041
Placebo (claims, falsification)	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	51	1.447

Notes: TWFE estimates restricted to state-quarters with non-missing T-MSIS outpatient claims data. Results consistent with full-panel estimates.

Table C6a. LPDID: Combined Channel

	Estimate	SE	t	p-value	2.5%	97.5%	N
treat_diff	-0.059	0.103	-0.568	0.573	-0.266	0.149	1,066

Table C6b. LPDID: Pharmacy Channel

	Estimate	SE	t	p-value	2.5%	97.5%	N
treat_diff	-0.022	0.066	-0.336	0.739	-0.155	0.111	1,066

Table C6c. LPDID: Claims Channel

	Estimate	SE	t	p-value	2.5%	97.5%	N
treat_diff	-0.029	0.086	-0.341	0.735	-0.202	0.143	582

Notes: Local projections difference-in-differences (Dube et al., 2023). Standard errors clustered at the state level.

Combined Panel Figures

Figure C1. Raw Trends in LARC Utilization by Channel, Treated vs. Never-Treated States

[Figure C1]

Notes: LARC utilization rates per 1,000 Medicaid enrollees by channel (pharmacy, outpatient procedures, combined). Solid lines = treated states; dashed lines = never-treated states. Trends are approximately parallel across all channels.

Figure C2. Event Study by Channel: LARC Utilization per 1,000 Enrollees

[Figure C2]

Notes: Event study coefficients and 95% confidence intervals from TWFE specifications by channel. Reference period is $k = -1$. Pre-treatment coefficients confirm parallel trends. Post-treatment effects are null across all channels.

Figure C3. Channel Comparison: Point Estimates and 95% CIs Across Estimators

[Figure C3]

Notes: TWFE, did2s, and LPDID point estimates with 95% confidence intervals for each channel. All estimates span zero, confirming the null result across channels and estimators.

Figure C4. Cohort-Specific Trends: Combined Panel

[Figure C4]

Notes: LARC utilization trends by adoption cohort in the combined panel.

Online Appendix

Postpartum Medicaid Extensions, Outpatient LARC Utilization, And The Medicaid Unwinding

Appendix A: Robustness And Sensitivity Analyses

APPENDIX EXHIBIT A1. Specification Curve: LARC TWFE Coefficients Across 14 Specifications

Specification	Coefficient	SE	p-value	N
Baseline TWFE	-0.010	0.091	0.915	1,836
Winsorized (95th pct)	0.031	0.086	0.721	1,836
Winsorized (99th pct)	-0.002	0.090	0.984	1,836
Log(LARC + 0.001)	0.037	0.291	0.900	1,836
Arcsinh(LARC)	0.013	0.061	0.828	1,836
Extensive margin	-0.000	0.039	0.995	1,836
Excl. COVID quarters	0.004	0.093	0.968	1,581

Specification	Coefficient	SE	p-value	N
Population-weighted	-0.040	0.067	0.559	1,836
Drop IL (early adopter)	-0.005	0.095	0.962	1,800
Drop late adopters (2024+)	-0.070	0.085	0.413	1,512
Large states only	-0.033	0.110	0.764	900
SPA adopters only	-0.022	0.095	0.814	1,764
State-specific trends	0.029	0.081	0.721	1,574
2018-2024 only	-0.012	0.080	0.878	1,428

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: All models include state and quarter fixed effects. Standard errors clustered at the state level. No specification yields statistical significance at the 0.05 level. The raw TWFE null is robust across all 14 alternative specifications, confirming that the null reflects the Medicaid unwinding confound rather than specification sensitivity.

APPENDIX EXHIBIT A2. Specification Curve Figure

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Each point represents the TWFE coefficient from one of 14 alternative specifications. Whiskers show 95 percent confidence intervals. All coefficients are centered near zero and none are statistically significant.

APPENDIX EXHIBIT A3. Randomization Inference Distribution

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Distribution of TWFE coefficients from 1,000 random permutations of treatment assignment. The observed coefficient (-0.010) is marked with a vertical line. Randomization inference p-value = 0.917, indicating the observed coefficient is well within the null distribution.

APPENDIX EXHIBIT A4. Leave-One-Out Jackknife Results (Selected States)

Dropped state	Coefficient	SE	p-value
Utah	-0.055	0.080	0.500
Wisconsin	0.034	0.085	0.689
Tennessee	-0.027	0.091	0.772
Alaska	-0.026	0.092	0.779
Florida	-0.021	0.095	0.826
Nevada	-0.036	0.091	0.696

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE coefficient re-estimated 51 times, each time dropping one state. Full range: [-0.055, 0.034]. No single state drives the raw null result. Full results available from the authors.

APPENDIX EXHIBIT A5. Leave-One-Out Sensitivity Figure

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE coefficients with 95 percent confidence intervals, each dropping one state. All estimates remain statistically insignificant.

Appendix B: Multi-Channel Analysis (SDUD + T-MSIS Outpatient Claims)

APPENDIX EXHIBIT B1. Descriptive Statistics: Combined Panel

Variable	Treated	Never-treated
N (state-quarters)	1,323	180
LARC per 1,000 (pharmacy)	0.954 (0.551)	0.722 (0.496)
LARC per 1,000 (outpatient claims)	0.422 (0.660)	0.377 (0.292)
LARC per 1,000 (combined)	1.242 (0.751)	1.015 (0.546)
LARC Rx count (pharmacy)	1,024 (1,145)	1,241 (890)
LARC claims (outpatient)	597 (2,693)	386 (307)
LARC volume (combined)	1,620 (3,288)	1,627 (985)
Total Medicaid enrollment	1,327,667 (1,847,956)	1,796,200 (1,142,406)
Claims channel share	0.161 (0.243)	0.274 (0.273)
Placebo per 1,000 (pharmacy)	0.550 (0.661)	0.571 (0.819)
Placebo per 1,000 (claims)	1.433 (1.595)	1.840 (1.357)
Placebo per 1,000 (combined)	1.529 (1.777)	2.002 (2.130)

SOURCE: Authors' analysis of SDUD, DOGE T-MSIS, and MBES data, 2016-2024. NOTES: Standard deviations in parentheses. Outpatient claims available for a subset of state-quarters (N=1,357 with non-missing data). Combined measure sums pharmacy and outpatient procedure counts.

APPENDIX EXHIBIT B2. TWFE Results By Channel

Channel	Coefficient	SE	p-value	95% CI	N	Mean
Combined (pharmacy + procedures)	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278

Channel	Coefficient	SE	p-value	95% CI	N	Mean
Combined (win-sorized)	-0.045	0.103	0.663	[-0.253, 0.162]	1,836	1.273
Outpatient procedures only	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403
Outpatient procedures (win-sorized)	-0.108	0.074	0.151	[-0.256, 0.041]	1,357	0.398
Pharmacy dispensing only	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy dispensing (win-sorized)	-0.002	0.090	0.984	[-0.183, 0.179]	1,836	0.975

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: All models include state and quarter fixed effects. Standard errors clustered at the state level. Outcomes expressed as LARC utilization per 1,000 total Medicaid enrollees.

APPENDIX EXHIBIT B3. did2s Results By Channel

Channel	Coefficient	SE	p-value	95% CI	N
Combined (pharmacy + procedures)	-0.076	0.111	0.495	[-0.293, 0.142]	1,836
Pharmacy dispensing only	-0.030	0.102	0.769	[-0.229, 0.169]	1,836

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: Gardner (2022) two-stage difference-in-differences. Standard errors clustered at the state level.

APPENDIX EXHIBIT B4. Heterogeneity: Combined Panel

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
Combined: All LARC	-0.055	0.105	0.604	[-0.265, 0.156]	1,836	1.278

Outcome	Coefficient	SE	p-value	95% CI	N	Mean
Pharmacy: All LARC	-0.010	0.091	0.915	[-0.192, 0.173]	1,836	0.980
Pharmacy: IUD	0.015	0.061	0.800	[-0.107, 0.137]	1,836	0.806
Pharmacy: Implant	-0.025	0.059	0.670	[-0.143, 0.093]	1,836	0.173
Pharmacy: Fee-for- service	-0.111	0.070	0.117	[-0.252, 0.029]	1,836	0.372
Pharmacy: Managed care	0.102	0.078	0.200	[-0.055, 0.258]	1,836	0.608
Claims: All LARC procedures	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	0.403
Placebo: Pharmacy	0.043	0.136	0.754	[-0.230, 0.315]	1,836	0.605
Placebo: Claims	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	1.447
Placebo: Combined	0.067	0.188	0.723	[-0.310, 0.444]	1,836	1.674

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: TWFE estimates with state and quarter fixed effects. Standard errors clustered at the state level. Placebo tests confirm no false positives in any channel.

APPENDIX EXHIBIT B5. Claims-Only Subsample Analysis (T-MSIS States Only)

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Claims: LARC proce- dures	-0.119	0.079	0.140	[-0.277, 0.040]	1,357	51	0.403
Combined: pharmacy + proce- dures	-0.127	0.108	0.248	[-0.344, 0.091]	1,428	51	1.424
Pharmacy only (T-MSIS subsam- ple)	-0.012	0.080	0.878	[-0.173, 0.149]	1,428	51	1.041

Outcome	Coefficient	SE	p-value	95% CI	N	N states	Mean
Placebo (claims, falsifica- tion)	-0.114	0.130	0.385	[-0.376, 0.148]	1,357	51	1.447

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: TWFE estimates restricted to state-quarters with non-missing T-MSIS outpatient claims data. Results consistent with full-panel estimates.

APPENDIX EXHIBIT B6. LPDID Estimates By Channel

Channel	Estimate	SE	p-value	95% CI	N
Combined	-0.059	0.103	0.573	[-0.266, 0.149]	1,066
Pharmacy	-0.022	0.066	0.739	[-0.155, 0.111]	1,066
Claims	-0.029	0.086	0.735	[-0.202, 0.143]	582

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: Local projections difference-in-differences (Dube et al., 2023). Standard errors clustered at the state level.

APPENDIX EXHIBIT B7. Raw Trends In LARC Utilization By Channel

[Figure]

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: LARC utilization rates per 1,000 Medicaid enrollees by channel (pharmacy, outpatient procedures, combined). Solid lines = treated states; dashed lines = never-treated states.

APPENDIX EXHIBIT B8. Event Study By Channel

[Figure]

SOURCE: Authors' analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: Event study coefficients and 95 percent confidence intervals from TWFE specifications by channel. Reference period is k=-1.

APPENDIX EXHIBIT B9. Channel Comparison: Point Estimates And CIs Across Estimators

[Figure]

SOURCE: Authors’ analysis of SDUD and DOGE T-MSIS data, 2016-2024. NOTES: TWFE, did2s, and LPDID point estimates with 95 percent confidence intervals for each channel. All estimates span zero in raw designs.

Appendix C: Power Analysis And Minimum Detectable Effects

APPENDIX EXHIBIT C1. Minimum Detectable Effects At 80 Percent Power

The minimum detectable effect (MDE) for the pharmacy channel is 0.255 per 1,000 total enrollees (26 percent of baseline). For the combined channel, the MDE is 0.293 per 1,000 (23 percent of baseline). Expressed per 1,000 postpartum women (assuming 2-3 percent postpartum share of total enrollment), MDEs range from 8.5 to 14.7.

Compared to literature benchmarks after denominator dilution: - Rodriguez et al. (2024) benchmark: 0.185 per 1,000 total enrollees (7.4 per 1,000 postpartum women). The pharmacy MDE of 0.255 **exceeds** this benchmark, meaning the raw TWFE design cannot detect effects of Rodriguez-scale magnitude. - Eliason et al. (2022) benchmark: 1.75 per 1,000 total enrollees (70.0 per 1,000 postpartum women). This is well within the detectable range.

The raw TWFE null is therefore partially uninformative: it can rule out very large effects but cannot distinguish between a true zero and a small-to-moderate positive effect.

Denominator Adjustment

Postpartum-eligible women represent approximately 1.9 percent of total Medicaid enrollment (scaling factor ~57x). The denominator-adjusted TWFE coefficient is -1.91 per 1,000 postpartum women (p=0.74, 95% CI [-13.5, 9.7]). The wide confidence intervals confirm that the raw design lacks power to detect clinically meaningful effects and help explain why the re-estimated DDD remains suggestive rather than decisive.

SOURCE: Authors’ calculations based on SDUD and MBES data. NOTES: MDEs computed at alpha=0.05, power=0.80, using the variance of the TWFE estimator. Denominator adjustment assumes postpartum-eligible women constitute approximately 1.9 percent of total Medicaid enrollment.

Appendix D: Dobbs Diagnostic

APPENDIX EXHIBIT D1. Heterogeneous Treatment Effects By State Abortion Policy Environment

Subgroup	Coefficient	SE	p-value
Hostile states	-0.058	0.135	0.670
Supportive states	0.067	0.101	0.510
Middle states	0.073	0.170	0.670
Post-Dobbs x hostile-state interaction	0.141	0.112	0.210

SOURCE: Authors’ analysis of SDUD data, 2016-2024. NOTES: TWFE estimates stratified by state abortion policy environment (hostile, middle, supportive) based on the Guttmacher Institute classification following the Dobbs v. Jackson Women’s Health Organization decision (June 2022). No significant differential effects by abortion policy environment, though the interaction estimate is imprecise.

Appendix E: Nexplanon NDC Compositional Break

APPENDIX EXHIBIT E1. Nexplanon NDC Transition Diagnostic

[Figure]

SOURCE: Authors’ analysis of SDUD data, 2016-2024. NOTES: Nexplanon (etonogestrel subdermal implant) received a new NDC labeler code in 2021 Q3 following Organon’s spinoff from Merck. SDUD data show zero implant prescriptions under Nexplanon NDCs prior to 2021 Q3, with a sharp discontinuity to positive counts beginning that quarter. This compositional break does not bias the DiD estimate unless it interacts differentially with treatment timing. The DDD results are not susceptible to this issue because the break is specific to LARC and does not affect negative control drugs.

Appendix F: Callaway-Sant’Anna Event Study

APPENDIX EXHIBIT F1. Callaway-Sant’Anna Event Study: LARC Prescriptions Per 1,000 Enrollees

[Figure]

SOURCE: Authors’ analysis of SDUD data, 2016-2024. NOTES: Callaway-Sant’Anna (2021) event study using both never-treated and not-yet-treated states as comparisons for each cohort-time cell. Pre-treatment ATTs are near zero ($t=-4$ through $t=-1$ all within 0.02 of zero), supporting the parallel trends assumption. Post-treatment ATTs are consistently positive, ranging from +0.038 ($t=0$) to +0.128 ($t=+6$), with an overall weighted post-treatment ATT of +0.052 ($SE=0.028$). Individual post-treatment coefficients are imprecise due to denominator dilution, but the consistently positive direction is compatible with a modest positive outpatient effect.

Appendix G: did2s Extended Drug Category Results

APPENDIX EXHIBIT G1. did2s Estimates For Extended Drug Categories

Drug category	Coefficient	SE	p-value
Short-acting contraceptives	-1.58	1.084	0.145
Antidepressants	-16.13	8.218	0.050
Antihypertensives	-26.93	9.961	0.007
Thyroid medications	-5.80	2.030	0.004
Diabetes medications	-7.99	3.043	0.009
Non-benzo anxiolytics	-5.06	6.165	0.412

Drug category	Coefficient	SE	p-value
Anticoagulants	-0.74	1.363	0.585
MOUD	-4.31	2.677	0.108
Statins (neg. control)	-13.87	3.963	0.001
Alzheimer's (neg. control)	0.13	0.080	0.109
BPH (neg. control)	-1.07	0.445	0.016
Gout (neg. control)	-0.84	0.342	0.015

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Gardner (2022) two-stage difference-in-differences estimates. Standard errors clustered at the state level. N=1,836 for all categories. Results corroborate the TWFE pattern: pervasive negative coefficients across all drug categories, including negative controls, confirming the Medicaid unwinding confound.

Appendix H: LARC DDD Robustness By Individual Negative Control

APPENDIX EXHIBIT H1. LARC-Specific DDD Against Individual Negative Controls

Specification	DDD coefficient	SE	p-value	95% CI
Pooled actual controls – TWFE (log1p)	+0.094	0.055	0.095	[-0.017, 0.204]
Pooled actual controls – did2s (log1p)	+0.109	0.062	0.078	[-0.012, 0.231]
Pooled actual controls – raw-rate sensitivity	+3.32	0.88	<0.001	[1.56, 5.09]
Pooled actual controls – asinh sensitivity	+0.105	0.069	0.135	[-0.034, 0.244]
Pooled actual controls – pre-period-index sensitivity	+0.079	0.097	0.423	[-0.117, 0.274]
Legacy pooled controls – raw-rate sensitivity	+12.20	3.53	0.001	[5.12, 19.28]
Statins only (log1p)	+0.187	0.064	0.005	[0.059, 0.314]
BPH medications only (log1p)	+0.094	0.058	0.114	[-0.023, 0.211]

Specification	DDD coefficient	SE	p-value	95% CI
Gout medications only (log1p)	+0.108	0.059	0.070	[-0.009, 0.226]
Alzheimer's drugs only (log1p)	-0.015	0.055	0.793	[-0.126, 0.096]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Live DDD estimates difference log-transformed LARC rates against log-transformed actual negative controls (statins, BPH medications, gout medications, Alzheimer's drugs). Raw-rate and alternative-normalization rows are sensitivity checks. The positive signal is concentrated in some specifications but not others, so the paper treats the DDD evidence as exploratory rather than definitive.

Appendix I: Additional Figures

APPENDIX EXHIBIT I1. Cohort-Specific Trends

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: LARC utilization trends by adoption cohort.

APPENDIX EXHIBIT I2. Multi-Drug TWFE Forest Plot

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: TWFE point estimates and 95 percent confidence intervals for all 12 drug categories. Nearly all categories show negative coefficients, including negative controls, revealing the Medicaid unwinding confound.

APPENDIX EXHIBIT I3. Drug Category Raw Trends

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Raw prescription rates per 1,000 enrollees by drug category, treated versus never-treated states.

APPENDIX EXHIBIT I4. Bacon Decomposition (Histogram)

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Goodman-Bacon (2021) decomposition of the TWFE estimator into timing-group comparisons. Weights and 2x2 DiD estimates for each comparison group pair.

APPENDIX EXHIBIT I5. Bacon Decomposition Scatter

[Figure]

SOURCE: Authors' analysis of SDUD data, 2016-2024. NOTES: Weight-by-estimate scatter plot of 1,900 pairwise 2x2 DD estimates from the Goodman-Bacon decomposition, color-coded by comparison type (treated vs. never-treated, early vs. late, late vs. already-treated). Already-treated-as-control comparisons carry 36 percent of total weight with mean estimate -0.041, revealing the contamination pattern that motivates the use of heterogeneity-robust estimators.

APPENDIX EXHIBIT I6. Treatment Timing Distribution

[Figure]

SOURCE: Authors' analysis of KFF and CMS policy data. NOTES: Bar chart showing the number of states adopting the 12-month postpartum extension per quarter, with a cumulative adoption curve. Forty-four of 51 jurisdictions adopted between 2021 Q3 and 2024 Q3 (mode: 9 states each in 2022 Q3 and 2022 Q4). Five jurisdictions remain never-treated. This figure is essential for assessing whether late dynamic effects (e.g., lags +5 to +8) are supported by adequate cohort sizes.

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